Graph Analytics

Prof. Laura Po

Some materials are taken from the book *Graph Databases in Action,* by Manning https://www.manning.com/books/graph-databases-in-action

Machine Learning & Graphs

OBJECTIVES

- An introduction to machine learning
- The role of graphs in machine learning applications

Machine Learning projects

Machine Learning (ML)

Delivering ML solutions is a complex process that requires more than selecting the right algorithm(s).

Such projects include numerous tasks related to [Sculley, 2015]

- ***Selecting*** the data sources
- ***Gathering*** data
- ***Understanding*** the data
- ***Cleaning*** and transforming the data
- Processing the data to ***create ML models***
- ***Evaluating*** the results
- ***Deploying***

After deployment, it is necessary to ***monitor*** the application and ***tune it***. The entire process involves multiple tools, a lot of data, and different people.

ML project life cycle: CRISP- DM

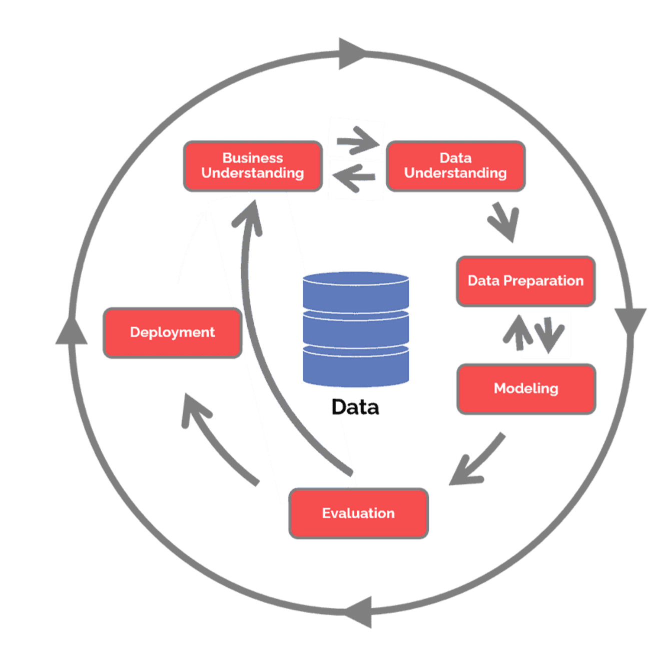
One of the most commonly used processes for data mining project s is the Cross-Industry Standard Process for Data Mining, or CRIS P-DM [Wirth and Hipp, 2000].

It can also be applied to generic machine learning projects.

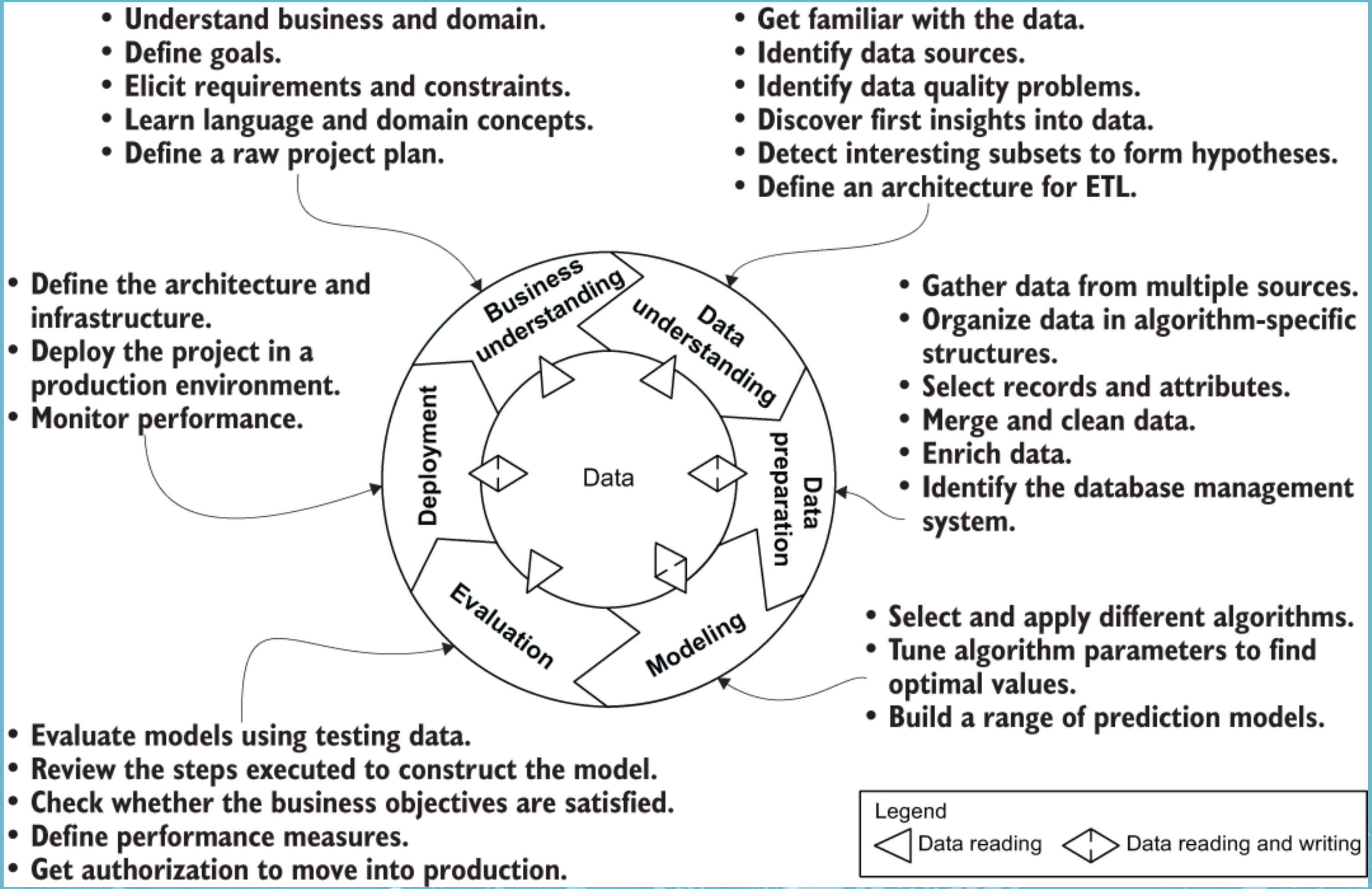
Key features of the CRISP-DM that make it attractive as part of the base workflow model are

- It is not proprietary.
- It is application, industry, and tool neutral.
- It explicitly views the data analytics process from both an applicat ion-focused and a technical perspective.

This method can be used for project planning and management, for communicating, and for documentation purposes.



https://hpi.de/fileadmin/user_upload/fachgebiete/rabl/Lectures/PDE_Poster/PDE_Patricia_Sowa.pdf



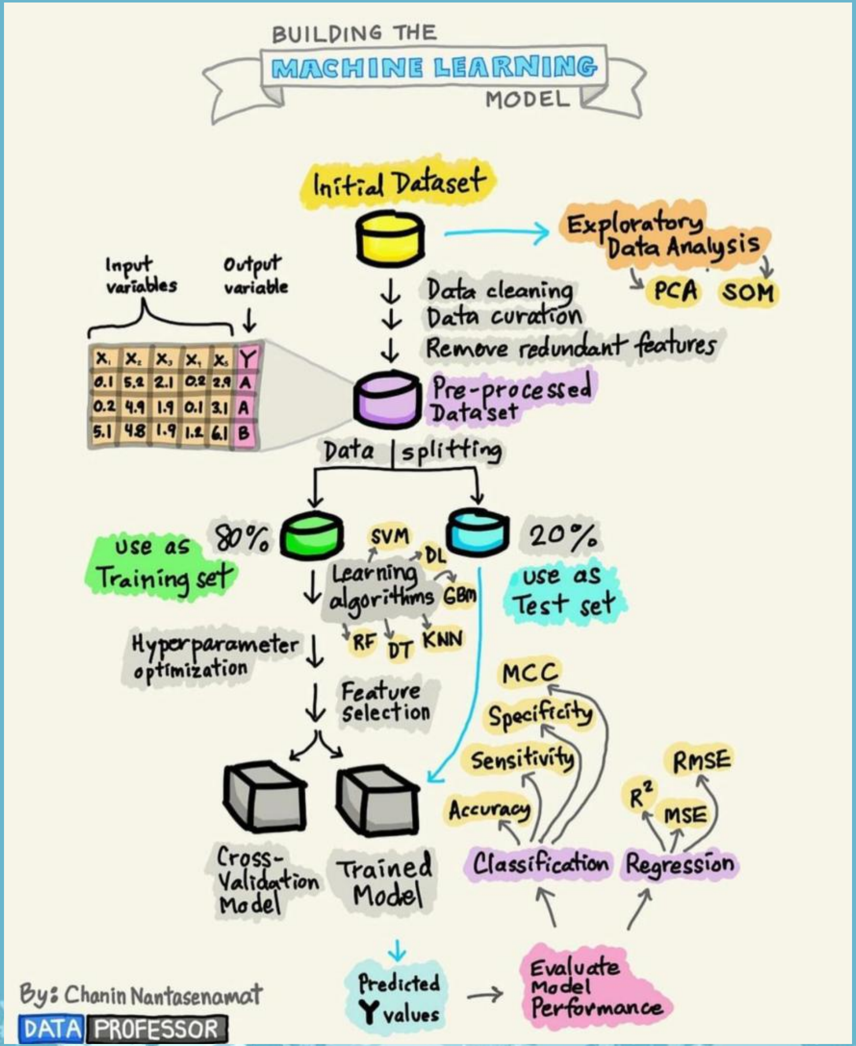
MORE DETAILS https://www.sv-europe.com/crisp-dm-methodology/

BUILDING THE MACHINE LEARNING MODEL

Initial Dataset

By: Chanin Nantasenamat

DATA PROFESSOR


BUILDING THE
MACHINE LEARNING
MODEL
Initial Dataset
Exploratory Data Analysis
PCA
SOM
Data cleaning
Data curation
Remove redundant features
Input variables
Output variable
X1 X2 X3 X4 X5 Y
0.1 5.2 2.1 0.2 2.9 A
0.2 4.9 1.9 0.1 3.1 A
5.1 4.8 1.9 1.2 6.1 B
Pre-processed Dataset
Data splitting
80%
Use as Training set
20%
Use as Test set
SVM
DL
Learning algorithms
GBM
RF
DT
KNN
Hyperparameter optimization
Feature Selection
Cross-Validation Model
Trained Model
MCC
Specificity
Sensitivity
Accuracy
Classification
RMSE
R2
MSE
Regression
Predicted Y values
Evaluate Model Performance
By: Chanin Nantasenamat
DATA PROFESSOR

Machine Learning challenges

The source of truth

“It’s an absolute myth that you can send an algorithm over raw data and have insights pop up.”

Jeffrey Heer, University of Washington

- The training data represents the ***source of truth*** from which any insight can be extracted and any pr ediction can be made. Managing the training requires a lot of effort.

- It has been estimated that data scientists spend ***up to 80%*** of their time on ***data preparation*** [Lohr, 2014]

Data - the first big challenge

- ***Insufficient quantity of data***—ML requires a lot of training data to work properly. Even for simple use c ases, it needs thousands of examples, and for complex problems such as deep learning or for nonlinear algorithms, you may need millions of examples.

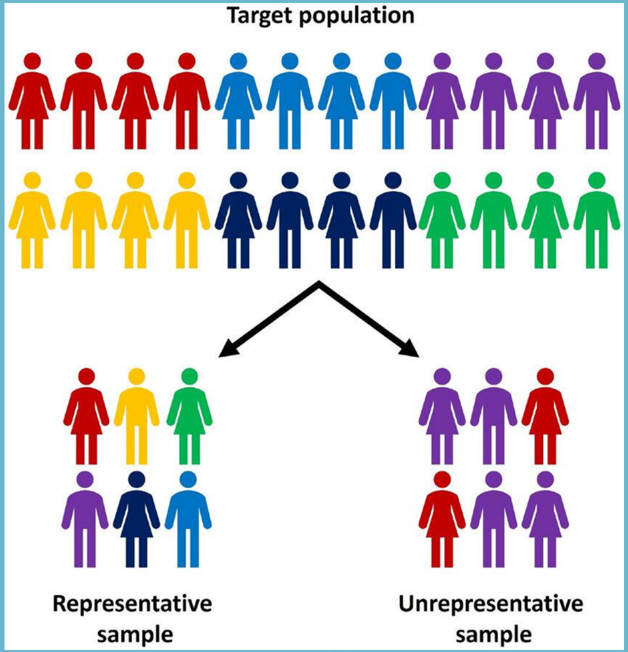
Data - the first big challenge

- ***Poor quality of data***—Data sources are always f ull of errors, outliers, and noise. Poor data quality directly affects the quality of the results of the ML process because it is hard for many algorithms to discard wrong (incorrect, unrelated, or irrelevant) v alues and to then detect underlying patterns.

Bad data **is an inaccurate set of information, including missing data, wrong information, ina ppropriate data, non-conforming data, duplica te data and poor entries (misspells, typos, vari ations in spellings, format etc).**

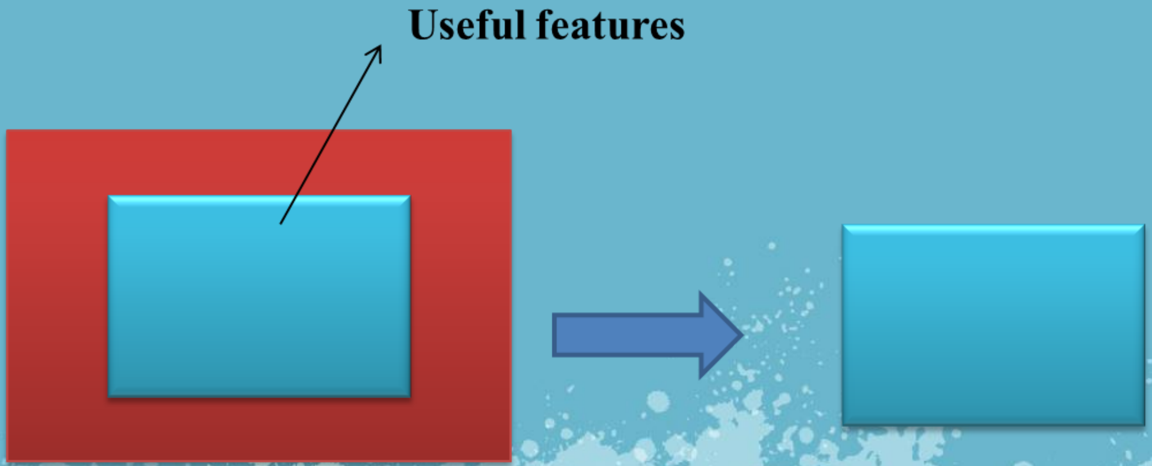
Data - the first big challenge

- ***Nonrepresentative data***— ML is a process of induction: the mo del makes inferences from what it has observed and will likely no t support cases that your training data does not include. If the trai ning data is too noisy or is related only to a subset of possible ca ses, the learner might generate bias or overfit the training data a nd will not be able to generalize to all the possible cases.



Data - the first big challenge

- ***Irrelevant features***—The algorithm will learn in the right way if the data contains a good set of relevant f eatures and not too many irrelevant features. Using more features will enable the learner to find a more detailed mapping from feature to target, which increases the risk that the model computed will overfit the data. Feature selection and feature extraction represent two important tasks.



Issues

The issues related to the ***quality of the training examples*** determine a set of data management constraints and requirements for the infrastructure of the ML project.

Managing big data —Gathering data from multiple data sources and merging it into a unified source of truth will generate a huge dataset; increasing the amount of (quality) data will improve the quality of the learning process.

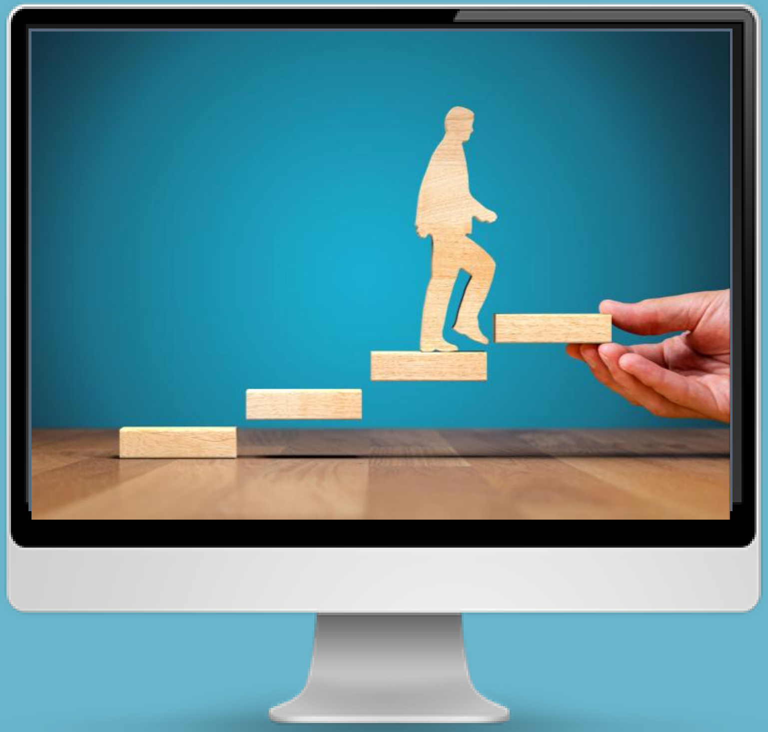
Issues

- **Designing a flexible schema** —Try to create a schema model that provides the capabilities to merge multiple heterogeneous schemas into a unified and homogeneous data structure that satisfies the informational and navigational requirements. The schema should evolve easily according to the project.

Issues

- **Developing efficient access patterns** —Fast data reads boost the performanc e, in terms of processing time, of the trai ning process. Tasks such as feature extr action, filtering, cleaning, merging, and other preprocessing of the training data will benefit from the use of a data platfor m that provides multiple and flexible acc ess patterns.

Performance



Performance is a complex topic in machine learning because it can be related to multipl e factors:

Predictive accuracy
Training performance
Prediction performance

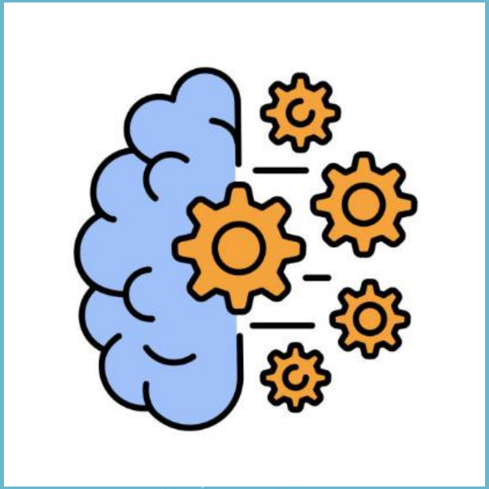
Performance

- ***Predictive accuracy***, which can be evaluated by using different performance measures. A typical performance measure for regression problems is the root mean squared error (RMSE), which measures the standard deviation of the errors the system makes in its predictions. It looks at the difference between the estimated value and the known value for all the samples in the test dataset and calculates the average value. In general, data plays a primary role in guaranteeing a proper level of accuracy.

Performance

- ***Training performance***, which refers to the time required to compute the model.
- The amount of data to be processed and the type of algorithm used determine the processing time and t he storage required for computing the prediction model.

- Clearly, this problem affects more the algorithms that produce a model as a result of the training phase. For instance-based learners, performance probl ems will appear later in the process, such as during prediction.
- In batch learning, the training time is generally longer, due to the amount of data to be processed (compared with the online learning approach, in whic h the algorithm learns incrementally from a smaller amount of data).
- Although in online learning, the amount of data to be processed is small, th e speed at which it is crunched affects the capacity of the system to be alig ned with the latest data available, which directly affects the accuracy of the predictions.



Performance

- ***Prediction performance***, which refers to the response time required to provide predictions.
- The output of the machine learning project could be a static one-time report to help managers make strategic decisions or an online service for end users. In the first case, the time required to complete the prediction phase and compute the model is not a significant concern, so long as the work is com pleted in a reasonable time frame. In the second case, prediction speed does matter, because it affe cts the user experience and the efficacy of the prediction.

The benefit of using Graphs

- These factors (*Predictive accuracy, Training performance, Prediction performance)* can be translated into multiple requirements for machine learning projects, such as fast access to the data source during training, high data quality, an efficient access pattern for the model to accelerate predictions, and so on.

In this context, graphs could provide the proper storage mechanism for both source and model data, reducing the access time required to read data as well as offering multiple algorithmic techniques for improving the accuracy of the predictions.

Storing the model

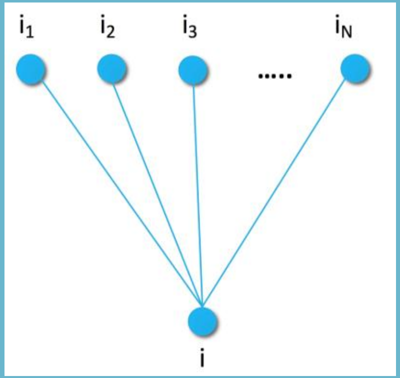
In the model-based learner approach, the output of the training phase is a model that will be used for making predictions.

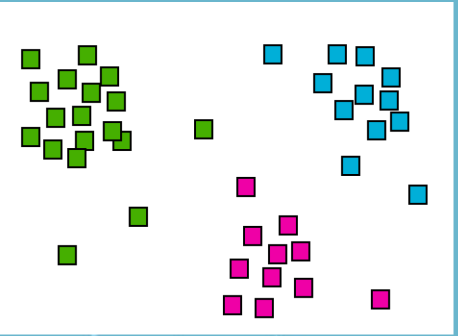
This model requires time to be computed and has to be stored in a persistence layer to avoid recomputation at every system restart.

The model's structure is related directly to the specific algorithm or the algorithm class employed.

Examples include

- **Item-to-item similarities** for a recommendation engine that uses a nearest-neighbor approach
- An **item-cluster mapping** that expresses how the elements are grouped in clusters





Storing the model

The sizes of the two models differ enormously. Consider a system that contains 100 items.

→**item-to-item similarities** would require 100 x 100 entries to be stored. Taking advantage of optimizations, this number can be reduced to consider only the top k similar items → 100 x k entries.

→**item-cluster mapping** requires only 100 entries; hence, the space required to store the model in memory or on disk could be huge or relatively modest. Moreover, model access/query time affects global performance during the prediction phase.

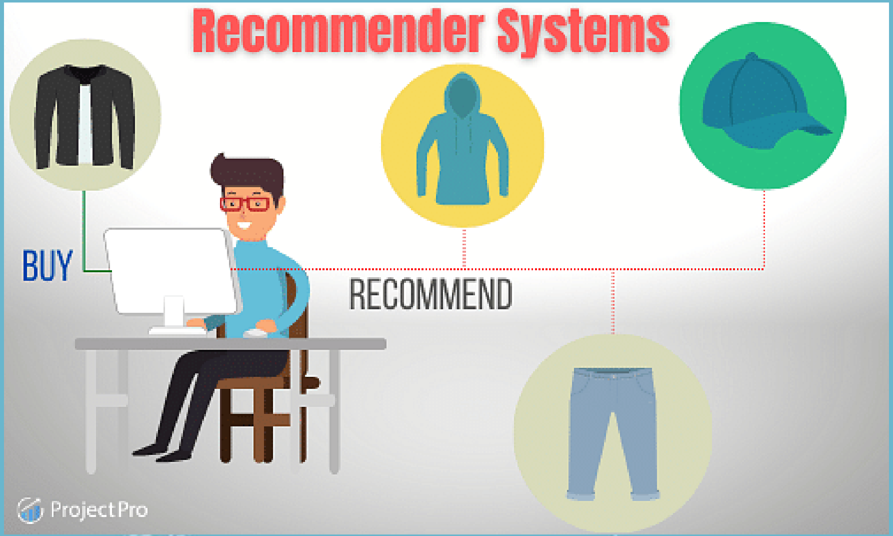
For these reasons, **model storage management** represents a significant challenge in machine learning.

Real time

Machine learning is being used more and more frequently to deliver real-time services to use rs.

Examples: simple recommendation engines that react to the user’s last clicks; self-driving cars that have been instructed not to injure a pedestrian crossing the street.

Although the outcomes in case of failure are vastly different, in both cases the capability o f the learner to **react quickly to new stimuli** coming from the environment is fundamental for the quality of the final results.

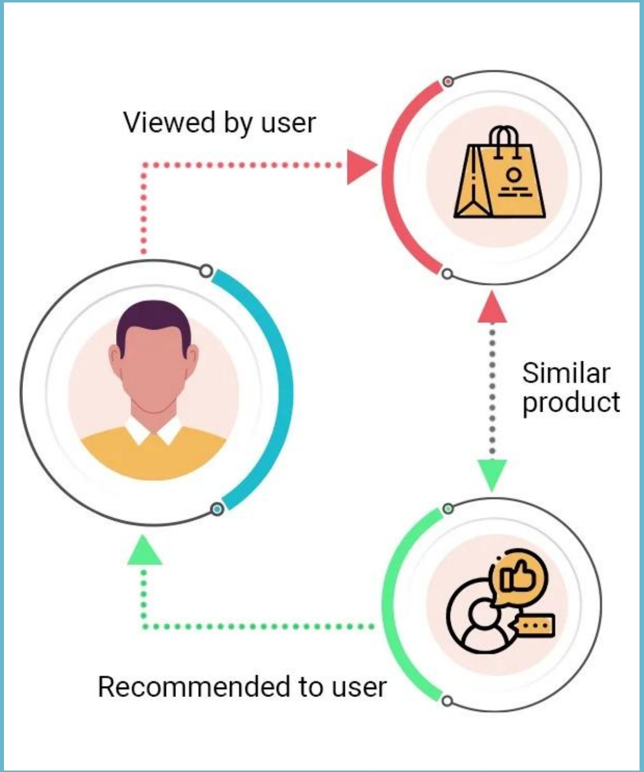


Real time

Consider a recommendation engine that provides real-time recommendations for anonymous users.

GOALS:

- ***Learn fast***. The online learner should be able to update the model as soon as new data is available. This capability will reduce the time gap between events or generic feedback, such as navigational clicks or interaction with a search session and updating of the model. The more the model is aligned to the latest events, the more it is able to meet the current needs of the user.
- ***Predict fast.*** When the model is updated, the prediction should be fast—a maximum of a few milliseconds—because the user may navigate away from the current page or even change their opinion quite rapidly.

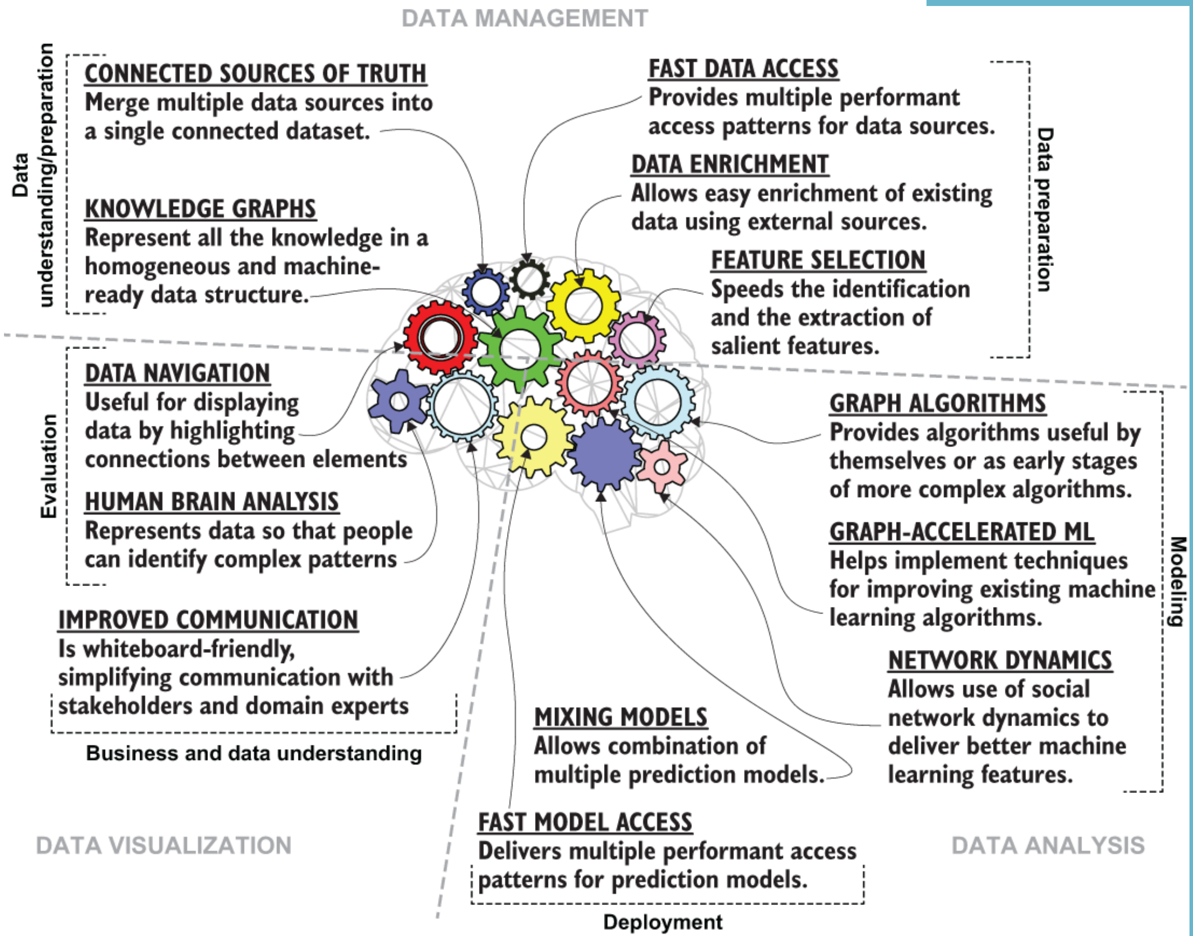


The role of graphs in machine learning

Graph-powered machine learning

- Graphs are used to characterize ***interactions*** between ***objects of interest***, to model simple and comp lex networks, or in general to represent real-world problems.
- Graph-based machine learning is becoming more common over time, transcending numerous tradition al techniques.

- Building a ***graph-powered machine learning platform*** has numerous benefits, because graphs can b e valuable also for delivering more advanced features that are impossible to implement without graph s upport.



Graph features are grouped into three main areas:

•***Data management***—This area c ontains the features provided by g raphs that help machine learning projects deal with the data.

•***Data analysis***—This area contai ns graph features and algorithms useful for learning and predicting.

•***Data visualization***—This area hi ghlights the utility of graphs as a v isual tool that helps people comm unicate, interact with data, and dis cover insights by using the human brain.

Data management

Graphs allow the learning system to explore more of the data, to access it faster, and to clean and enrich i t easily. Traditional learning systems train on a single table prepared by the researcher, whereas a graph-native system can access more than that table.

Graph-powered data management features include

- ***Connected sources of truth***—Graphs allow to merge multiple data sources into a single connected da taset ready for the training phase. It reduces data sparsity, increases the amount of data available, and simplifies data management.

- ***Knowledge graphs***— Knowledge graphs provide a homogeneous data structure for combining not onl y data sources, but also prediction models, manually provided data, and external sources of knowledge. The resulting data is machine ready and can be used during training, prediction, or visualization.

…

Data management (2)

Graph-powered data management features include

…

- ***Fast data access***—Tables provide a single access pattern related to row and column filters. Graphs provide multiple access points to the same set of data. This feature improves performance by reducing the amount of data to be accessed for a specific set of needs.

- ***Data enrichment***—Make it easy to extend existing data with external sources. The schemaless nature of graphs and the access patterns provided within graph databases help with data cleaning and merging.

- ***Feature selection***—Identifying relevant features in a dataset is key in several machine learning tasks, such as classification. By providing fast access to data and multiple query patterns, graphs speed feature identification and extraction.

Data analysis

Graphs can be used to model and analyze the relationships between entities as well as their properties. The schema flexibility provided by graphs also allows different models to coexist in the same dataset.

Graph-powered data analysis features include

- ***Graph algorithms***—Several types of graph algorithms, such as clustering, page ranking, and link analysis algorithms, are useful for identifying insights in the data and for analysis purposes. Moreover, they can be used as a first data preprocessing step in a more complex analysis process.

- ***Graph-accelerated machine learning***—The graph-powered feature extraction is an example of how graphs can speed or improve the quality of the learning system. Graphs can help in filtering, cleaning, enriching, and merging data before or during training phases.

…

Data analysis

Graph-powered data analysis features include

…

- ***Network dynamics***—Awareness of the surrounding contexts and related forces that act on networks allows not only to understand network dynamics, but also to use them to improve the quality of the predictions.

- ***Mixing models***—Multiple models can coexist in the same graph, taking advantage of flexible and fast access patterns, provided that they can be merged during the prediction phase. This feature improves final accuracy. Moreover, the same model sometimes can be used in different ways.

- ***Fast model access***—Real-time use requires fast predictions, which implies a model that can be accessed as fast as possible. Graphs provide the right patterns for these scopes.

Data visualization

Graphs have high communication power, and they can display multiple types of information at the same time in a way the human brain can easily understand.

Graph-powered data visualization features include

- ***Data navigation***—Networks are useful for displaying data by highlighting connections between elements. They can be used both as aids to help people navigate the data properly and as powerful investigation tools.

- ***Human-brain analysis***—Displaying data in the form of a graph unleashes the power of machine learning by combining it with the power of the human brain, enabling efficient, advanced, and sophisticated data processing and pattern recognition.

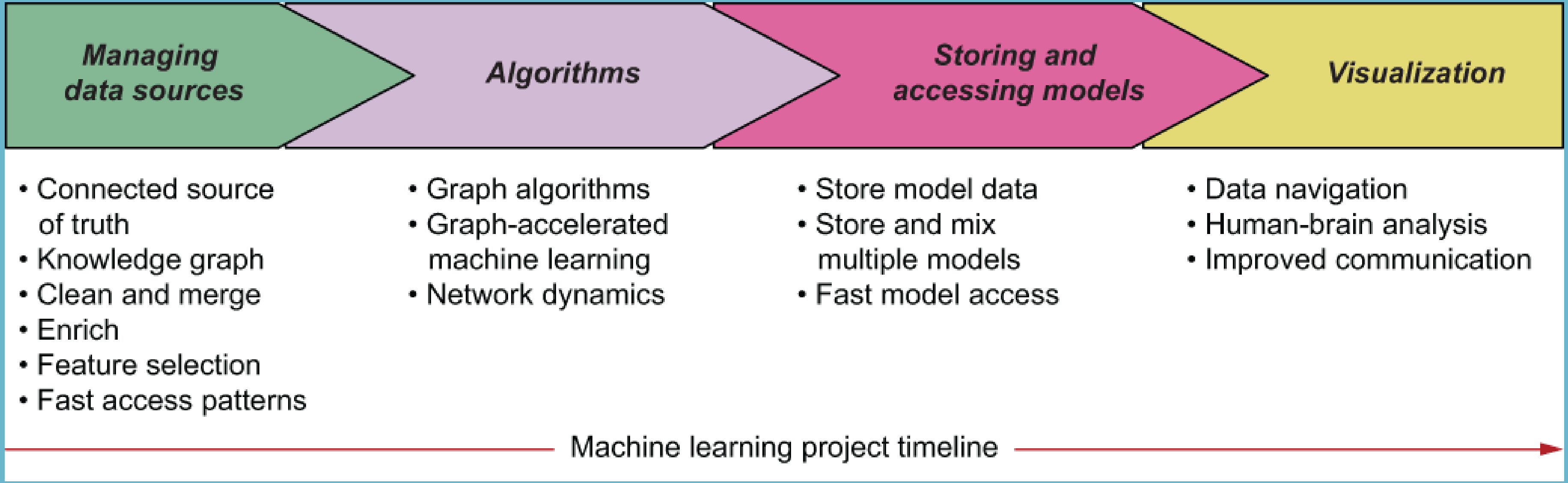
- ***Improved communication***—Graphs—in particular, property graphs—are “whiteboard friendly,” which means they are conceptually represented on a board as they are stored in the database. This feature reduces the gap between the technicalities of a complex model and the way in which it is communicated to the domain experts or stakeholders.

Graphs in 4 stages

The role of graphs depends on each specific case. Not in all machine learning projects we use graphs for all data management, data analysis and visualization.

We can use graphs in different ways in each of the ***four stages of a machine learning workflow***:

- *Managing the data sources*
- *Learning*,
- *Storing and accessing the model*,
- *Visualizing*.


Managing data sources
Algorithms
Storing and accessing models
Visualization
• Connected source of truth
• Knowledge graph
• Clean and merge
• Enrich
• Feature selection
• Fast access patterns
• Graph algorithms
• Graph-accelerated machine learning
• Network dynamics
• Store model data
• Store and mix multiple models
• Fast model access
• Data navigation
• Human-brain analysis
• Improved communication
Machine learning project timeline

Thank you

Insert the title of your subtitle Here